Technical & Product Documentation: KDIA Re Park Customer Portal

Application Name: KDIA Re Park Customer Portal
Version: 1.0.0 (MVP Phase)
Date: January 04, 2026
Document Status: Authority Reference

1. Title Page

Document Title: Comprehensive Technical & Product Reference
Application Name: KDIA Re Park Customer Portal
Version / Phase: v1.0 / MVP Baseline
Date: 2026-01-04
Purpose: This document provides a detailed technical and functional overview of the KDIA Re Park **Customer Portal**. It serves as the primary reference for internal stakeholders, developers, and management regarding implemented features, system architecture, and design philosophy.

2. Application Overview

The KDIA Re Park Customer Portal is a centralized web application designed for consumers enrolled under the Energy Investment Plan (EIP). It allows users to manage their virtual energy allocations, track infrastructure performance, and monitor estimated clean energy credits.

Target Audience: Residential and SME consumers participating in the KDIA Re Park renewable initiative.
Problem Solved: Bridges the gap between centralized solar infrastructure and distributed consumption by providing a transparent ledger for virtual energy offsets without requiring rooftop installations.
Scope of MVP: Focuses on secure authentication, EIP-aligned allocation selection, and an "Infrastructure-grade" dashboard for consumption monitoring.

3. High-Level Architecture

The portal follows a standard modern web architecture designed for modularity and security.

Architecture Pattern: Client–Server (Decoupled).
Frontend Stack:
React (Vite): Comprehensive UI components and state management.
Tailwind CSS: Professional styling via a custom design system.

Recharts: Data visualization for energy performance history.
Backend Stack:
Node.js & Express: RESTful API services.
JSON Web Tokens (JWT): Stateless session handling.
Bcrypt: Industry-standard password hashing.
Database:
SQLite: Chosen for the MVP phase to ensure portability and ease of demonstration while maintaining SQL compliance.
Security Approach: Implementation of authenticated route guards, secure password storage, and environment-based configuration.

4. User Roles & Access

Role: Customer
Description: Individual or business entity with a valid utility Consumer ID.
Permissions: Can register, select an energy allocation segment, view personal consumption totals, and access performance ledgers.
Authentication & Session handling: Users remain authenticated via a JWT stored securely (memory/local). Sessions are non-persistent by default for the MVP.
Access Boundaries: Verified customers can only access data tied to their specific Consumer ID and Subscription ID.

5. Authentication Features

Registration Flow
Required Fields: Full Name, Email/Mobile, Utility Consumer ID, and Security Password.
Auto-generated Consumer ID: The system validates existing IDs or simulates the initialization of a new KDIA-tracked identity upon first entry.
Password Rules: Strict validation enforcing 8+ characters, uppercase, lowercase, digit, and special character.
UI Enhancements: Centered premium cards with dynamic validation checklists and trust statements ("Protected Infrastructure").
Login Flow
Identifier: Email or Mobile number.
Security: Rate-limiting and error-masking (generic "Invalid credentials" messages) to prevent account enumeration.
UI Enhancements: Professional, minimalist design focusing on focus states (input-premium) and brand trust.

6. Subscription & Energy Allocation Features

The portal utilizes the Energy Allocation Plan concept, moving away from retail "subscriptions" to align with state infrastructure frameworks.

Plan Types (Energy Segments)
Small Energy Allocation (500 kWh): Tailored for apartments and low-load users (~75% coverage).
Medium Energy Allocation (1,000 kWh): Optimized for standard households (~100% coverage).
Large Energy Allocation (2,000 kWh): Designed for high-demand residences or SMEs (100%+ capacity).
Indicative Nature: All plans are presented as "Indicative" based on infrastructure projections.
Selection Flow: Users select a segment based on their consumption profile, which records their "Reserve Intent" in the backend.
Redirection Logic: Post-selection, users are automatically initialized and routed to the secure Dashboard.

7. Dashboard Features (Detailed)

The Dashboard serves as the operational headquarters for the customer's green energy portfolio.

Hero Section: Personalized greeting with high-level operational metadata (Subscription ID, Status, Start Date).
Subscription Summary: Clear display of total allocated units vs. current period status.
Energy Consumption Tracking: Real-time (simulated) monitoring of consumed vs. available units.
Progress Indicators: High-contrast radial and linear meters showing the percentage of allocation utilized.
Historical Charts: Interactive bar charts showing the 6-month historical utilization trend.

8. Value-Added Insights (Mock / Computed)

The MVP incorporates advanced insights to demonstrate long-term value.
Estimated Savings: Projected financial offsets based on grid tariff assumptions.
Environmental Impact: Real-time conversion of energy usage into CO_2 kg offset and "Trees Planted" equivalents.
Usage Patterns: Insights identifying high-efficiency periods (e.g., "Operational Insight: Night Load Optimization").
Billing & Adjustment Ledger: A transparent view of utilized units vs. base allocation.
Activity Timeline: Log of infrastructure events (e.g., "Allocation Synced," "Statement Generated").
NOTE

Data Disclaimer: All financial metrics and environmental impact values are Estimated / Mock – computed on the frontend for demonstration purposes.

9. UI & UX Design Philosophy

The portal adheres to an Infrastructure-grade design philosophy, prioritizing stability and trustworthiness over flashy visuals.

Color Palette:
Primary: Deep Teal (#0D9488) symbolizing sustainability and energy.
Secondary: Slate and Neutral tones for a professional, enterprise feel.
Typography: Modern inter-family fonts with strategic use of "Black" weights for infrastructure headers.
Components:
card-premium: High-radius (2.5rem), soft-shadow containers.
btn-premium: High-contrast, tactile buttons with subtle scale transforms.
Micro-interactions: Smooth Tailwind-based transitions for hover states, scale transforms on interaction, and subtle loading pulses.

10. Educational & Trust-Building Elements

Framework Education: A dedicated section explaining Virtual Net Metering (VNM) and Group Net Metering (GNM) to ensure users understand the regulatory backbone.
Tooltips: Informative hovers on complex abbreviations (e.g., kWh, EIP).
Disclaimers: Ubiquitous "Indicative" tagging on all savings and yield calculations to maintain regulatory transparency.

11. Empty States & Edge Case Handling

New Users: Guided flows for initial registration and plan selection.
No Data States: Professional placeholders for usage history (e.g., "Awaiting first cycle performance data").
Loading States: Skeleton-style pulses and progress bars to maintain perceived performance.
Error Messaging: User-friendly alerts with actionable next steps (e.g., "Check Consumer ID format").

12. Verification & Testing

Manual Verification: End-to-end traversal of the Register → Subscribe → Dashboard flow.
UI Validation: Cross-browser testing for responsive alignment on Mobile, Tablet (iPad Pro), and Desktop (1080p+).
Flow Validation: Ensuring authentication guards prevent unauthorized access to /dashboard or /subscription routes.

13. Limitations of Current MVP

As a pre-production MVP, the following limitations apply:

Mock Data: Consumption and savings data are currently generated via client-side simulation.
No Payment Gateway: Onboarding is currently an "Intent" reservation only.
No Admin Panel: User management is limited to internal database access.
No Real-time DISCOM Sync: Data relies on simulated API responses rather than live utility endpoints.

14. Future Roadmap (High Level)

Payment Integration: Direct processing of monthly EIP contributions.
Admin Dashboard: Comprehensive reporting for KDIA management.
PDF Statement Engine: Automated generation of official allocation ledgers.
Notification Hub: Email/SMS alerts for low balance and infrastructure outages.
Regulatory API Integration: Live synchronization with state DISCOM billing systems.

15. Conclusion

The KDIA Re Park Customer Portal v1.0 successfully establishes a premium digital presence for the Energy Investment Plan. It provides a robust, secure, and visually cohesive foundation for managing centralized clean energy allocations. The application is currently ready for high-level stakeholder demonstrations and internal validation of the user experience.